Rough Heston with jumps - joint calibration to SPX/VIX level and skew as $T \to 0$

Martin Forde Benjamin Smith*

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Abstract

We augment the well-known rough Heston model with an additional independent CGMY-type Lévy process with a Brownian component. We show that, subject to parameter feasibility conditions, one can simultaneously use the rough Heston parameters to fit the at-the-money VIX level and skew as $T \to 0$, and the CGMY parameters to fit the observed level, at-the-money correction and at-the-money skew of SPX options as $T \to 0$ (using the main Theorem in [FSV21] adapted for our rough Heston V process). The drift of the V process can also be made consistent with the initial observed variance curve structure, provided the resulting curve is admissible. In this sense, the additional Lévy component allows the SPX and VIX smiles to decouple as $T \to 0$ (which is not possible for a rough model without jumps), and the model can generate power-law skew for both smiles with different effective H-values.

1 Introduction

The Rough Heston stochastic volatility model was introduced by Jaisson and Rosenbaum [JR16], and (using C-tightness arguments from Jacod and Shiryaev [JS13]) they show that the model arises naturally as a weak large-time limit of a high-frequency market microstructure model driven by two nearly unstable Hawkes processes. [ER19] show that the characteristic function of the log stock price for the Rough Heston model admits a quasi-closed form solution via the solution to a non-linear Volterra integral equation (VIE) (see also [ER18]), and the variance curve for the model evolves as $d\xi_t(u) = \kappa(u-t)\sqrt{V_t}dW_t$, where $\kappa(t)$ is the usual fractional kernel $t^{H-\frac{1}{2}}$ for the V process multiplied by a *Mittag-Leffler* function. The instantaneous variance process V for the model is $(H-\varepsilon)$-Hölder continuous like fractional Brownian motion (see e.g. Theorem 3.2 in [JR16]) and the model exhibits power law skew in the small-time limit (see Theorem 3.1 in [FGS21] and Corollary 3.4 in [FSV21]). The theoretical value of the VIX index at time t satisfies $\text{VIX}_t^2 = -\frac{2}{\Delta}\mathbb{E}^{\mathbb{Q}}(\log(e^{-(r-q)\Delta}\frac{S_{t+\Delta}}{S_t})|\mathcal{F}_t)$, where S_t is the S&P 500 index value at time t, $\Delta = 30$ days and $\mathcal{F}_t$ is the market filtration, so VIX_t is effectively a rolling 30-day option-implied, log-contract-based variance rate. With jumps this convention need not coincide with the expectation of realised quadratic variation. A VIX option is a European call or put option on VIX_T for some maturity T, and if we replace the spot value S_0 in the Black-Scholes formula with the VIX future price $\mathbb{E}(\text{VIX}_T)$, we can define the implied volatility of a VIX call or put in the usual way by inverting the Black-Scholes formula. VIX options are very liquid in practice (although their bid/offer spreads are still comparatively high), and empirical and model-generated VIX implied volatility smiles have always exhibited a marked positive skew.

In [FGS21b], we consider a generalized Rough Heston model with non-zero initial variance curve $\xi_0(t)$, and we show that (modulo a scaling factor) $(\text{VIX}_T^2 - \text{VIX}_0^2)/T^{\frac{1}{2}-H}$ satisfies the same small-time LDP as the re-scaled log stock price $(X_T - X_0)/T^{\frac{1}{2}-H}$ in the main Theorem 3.3 in [FGS21] if we set $\rho = 1$. We later translate this LDP into VIX call option and implied volatility asymptotics, and we compute a small log-moneyness expansion for the asymptotic VIX smile using expansions previously derived in [FGS21], yielding tractable expressions for the overall level, skew and convexity of the short-end VIX smile.

Unfortunately, since the limiting VIX smile only depends on the factor $\nu/\sqrt{V_0}$ and not on ρ, we cannot simultaneously fit the overall level and skew of observed limiting VIX smile using the standard rough Heston model. To circumvent this issue, we enrich the model in this article with an additional independent CGMY (a.k.a. KoBoL)-type Lévy process L as in [FSV21] with $Y \in (1,2)$ (which implies L has infinite variation), and using a simple modification of the main result in [FSV21] for the Edgeworth regime where log-moneyness scales like $x\sqrt{T}$, we show that the SPX and VIX smiles decouple in some sense as $T \to 0$ if $1 - Y/2 < H$.

More specifically, we can use the rough Heston parameters H, V_0, ν and the initial value VIX_0 of the VIX to fit the overall level and the at-the-money skew of VIX options as $T \to 0$. We can then use the Brownian component σ and the CGMY jump parameters Y, C_+ and C_- to fit the observed at-the-money implied volatility as $T \to 0$, the first order at-the-money correction and the at-the-money skew of SPX options as $T \to 0$. Finally we choose the CGMY parameters G and/or M and $\xi_0(t) = \mathbb{E}(V_t)$ in the definition of the V process to be consistent with the observed variance term structure and the value that we have just calibrated for V_0. The upshot of this is that our CGMY-rough Heston model can generate different power-law skew for the SPX and VIX smiles in the $T \to 0$ limit.

*Dept. Mathematics, King's College London, Strand, London, WC2R 2LS (Benjamin.Smith@kcl.ac.uk)

2 The model

Consider a generalized Rough Heston model for a log stock price process $X_t = \log S_t$:

$$\left\{ \begin{array}{rcl} dX_t &=& -\frac{1}{2}V_t dt + \sqrt{V_t}(\rho dW_t + \bar{\rho} dB_t)\,, \\ V_t &=& \xi_0(t) + c_\alpha \int_0^t (t-s)^{\alpha-1} \nu \sqrt{V_s} dW_s \end{array} \right. \tag{1}$$

for $H \in (0, \frac{1}{2})$, $\alpha = H + \frac{1}{2}$, $c_\alpha = \frac{1}{\Gamma(\alpha)}$ and $\nu > 0$, with some nonnegative initial variance curve $\xi_0(t)$ with $\xi_0(.)$ continuous and $\xi_0(0) = V_0 > 0$, for which the square-root Volterra equation admits a nonnegative solution, where W, B are two independent Brownian motions, $\bar{\rho} = \sqrt{1-\rho^2}$ with $|\rho| \le 1$, and we assume $X_0 = 0$ and zero interest rate without loss of generality. It is not known whether we have pathwise uniqueness for (1) even when $\xi_0(t)$ is constant because $\sqrt{v}$ is not Lipschitz at zero (see section 4.2.3 in [JP20] for more on this), but we do have weak uniqueness (see Theorem 3.4 in [ALP19]) and uniqueness in law for V on $C([0,T])$, since we can explicitly compute an exponential-affine formula for the Fourier transform of V on pathspace in terms of a Volterra integral equation with a unique solution, see Theorem 7.1 in [ALP19] (see also Theorem 6.1 there); the fixed-t version of this formula is used in Appendix B.

2.1 Adding jumps - decoupling the SPX and VIX short-maturity skews

We now augment our rough Heston model to include jumps of infinite variation. Specifically (following [FSV21]), we now assume that the log stock price process X evolves as

$$dX_t \;=\; -\frac{1}{2}V_t dt + \sqrt{V_t}(\rho dW_t + \bar{\rho} dB_t) - dL_t \tag{2}$$

where L is a Lévy process independent of W and B with $\mathbb{E}(e^{-uL_t}) = e^{tV(u)}$ for u in some interval (u_-, u_+) which includes 0 and 1 with $\mathbb{E}(e^{-L_t}) = 1$ (i.e. $V(1) = 0$) which ensures that e^{-L} is a martingale with respect to its own filtration, and we assume that $X_0 = 0$ and impose the nondegeneracy assumption that the Lévy cumulant $V(u)$ is not identically zero.

Then

$$-\frac{2}{\Delta}\mathbb{E}(X_{T+\Delta} - X_T|\mathcal{F}_T) \;=\; \frac{1}{\Delta}\mathbb{E}(\int_T^{T+\Delta} V_u du|\mathcal{F}_T) \,-\, \frac{2}{\Delta}\mathbb{E}(-(L_{T+\Delta} - L_T)|\mathcal{F}_T) \;=\; \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u) du \,+\, a_1$$

where $\xi_t(u) := \mathbb{E}(V_u|\mathcal{F}_t)$, and

$$a_1 \;:=\; -2V'(0) \;=\; -\frac{2}{\Delta}\mathbb{E}(-(L_{T+\Delta} - L_T)|\mathcal{F}_T) \;\ge\; -\frac{2}{\Delta}\log \mathbb{E}(e^{-(L_{T+\Delta} - L_T)}|\mathcal{F}_T) \;=\; -\frac{2}{\Delta}\log 1 \;=\; 0 \tag{3}$$

and we have used Jensen and the martingale condition to obtain the inequality here. Moreover, since $V(0) = V(1) = 0$, $V'(0) < 0$ (since the Lévy cumulant $V(u)$ is not identically zero), so we see that a_1 is strictly positive.

Then we see that $\mathrm{VIX}_T^2 = \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u) du + a_1$, so the effect of the independent Lévy component is to increase the value of VIX_T^2 by a *shift factor*[1] equal to a_1. Thus $(\mathrm{VIX}_T^2 - a_1 - \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du)/T^{\frac{1}{2}-H} = (\mathrm{VIX}_T^2 - \mathrm{VIX}_0^2)/T^{\frac{1}{2}-H}$ now satisfies the LDP as $T \to 0$ with speed T^{-2H} and rate function $J(x)$ as before, and we have the following:

Corollary 2.1

$$\begin{aligned} \lim_{T\to 0} T^{2H}\log \mathbb{E}((\mathrm{VIX}_T - (a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du)^{\frac{1}{2}}\, e^{xT^{\frac{1}{2}-H}})_+) \;&=\; \lim_{T\to 0} T^{2H}\log \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\, e^{xT^{\frac{1}{2}-H}})_+) \\ &=\; -J(2(a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du)x) \;=\; -J(2\mathrm{VIX}_0^2 x) \end{aligned}$$

for $x > 0$, where $J(.)$ is the same as in the main Theorem 2.1 in [FGS21b].

Proof. See Appendix A. ■

Corollary 2.2

$$\hat{\sigma}_{\mathrm{VIX}}(x) \;:=\; \lim_{T\to 0} \hat{\sigma}_{\mathrm{VIX}}(\mathrm{VIX}_0 e^{xT^{\frac{1}{2}-H}}, T) \;=\; \frac{|x|}{\sqrt{2J(2\mathrm{VIX}_0^2 x)}}\,. \tag{4}$$

for $x > 0$. The value at $x = 0$ is understood as the continuous limit as $x \downarrow 0$; the corresponding formula for $x < 0$ requires the out-of-the-money put asymptotics.

[1] Note this is conceptually similar to a simple displaced-diffusion model $dS_t = (\beta S_t + 1 - \beta)dW_t$ for S with $\beta \in (0,1)$ where S can go negative, but here we do not have this negativity problem since $a_1 \ge 0$. This also means that a VIX put option with $K \le \sqrt{a_1}$ is worthless.

Expanding the final expression in Corollary 2.2 as $x \downarrow 0$, we find that

$$\begin{aligned}\hat{\sigma}_{\mathrm{VIX}}(x) &= \frac{\nu\sqrt{V_0}}{\frac{1}{\Delta}\int_0^\Delta \xi_0(u)du + a_1}\,\frac{\Delta^{\alpha-1}}{2\alpha\Gamma(\alpha)} + \frac{\nu}{2\sqrt{V_0}\Gamma(2+\alpha)}x + O(x^2) \\ &= \frac{\nu\sqrt{V_0}}{\mathrm{VIX}_0^2}\,\frac{\Delta^{\alpha-1}}{2\alpha\Gamma(\alpha)} + \frac{\nu}{2\sqrt{V_0}\Gamma(2+\alpha)}x + O(x^2)\,. \end{aligned} \tag{5}$$

Remark 2.1 Since $a_1 \geq 0$, we see that if the leading order term is fixed, adding jumps can only increase the skew term relative to the leading order term.

2.2 CGMY jumps and asymptotics for the SPX skew as $T \to 0$ in the Edgeworth regime

From here on we consider the special case when L is a generalized tempered stable (i.e. CGMY-type) process with $L_0 = 0$, Brownian variance σ^2, and Lévy density $\nu(x) = \frac{C_+e^{-Mx}}{x^{1+Y}}1_{x>0} + \frac{C_-e^{-G|x|}}{|x|^{1+Y}}1_{x<0}$ with $Y \in (1,2)$, which implies that

$$\phi_t^L(u) := \mathbb{E}(e^{-iu(L_t-L_0)}) = e^{t\psi(u)} = e^{iubt - \frac{1}{2}\sigma^2u^2t + C_+\Gamma(-Y)t[(M+iu)^Y - M^Y] + C_-\Gamma(-Y)t[(G-iu)^Y - G^Y]} \tag{6}$$

and $\mathbb{E}(e^{-pL_t}) = e^{V(p)t}$ for $p \in (-M, G)$, where $V(p) = \psi(-ip)$. Here b is the linear coefficient in the characteristic exponent of $-L$, not the drift of L in the standard Lévy–Khintchine representation. For example, using the truncation function $h(x) = x$, the drift of L is $-b - Y\Gamma(-Y)(C_+M^{Y-1} - C_-G^{Y-1})$. Note that b is fixed by the martingale condition $V(1) = 0$.

Remark 2.2 Corollary 2.10 and Remark 2.7 in [FSV21] describe a method for simulating L_t by changing to a measure under which L is a pure α-stable process, which allows for long-time stepping using the classical Chambers, Mallows and Stuck [CMS76] formula for sampling stable random variables.

The bivariate weak convergence used for the rough-volatility correction in the main Theorem 2.1 of [FSV21] is established for our rough Heston model in Appendix B: the Gaussian approximation $X_t^{(0)}$ there plays the role of $\hat{X}_t$ in Eq. 17 of [FSV21]. For the leading CGMY correction, however, a quantitative comparison with the constant-volatility model suffices, together with the localisation in Appendix C and the pure-jump expansion in [FSV21]; the independent Brownian component of L is incorporated in the leading Gaussian variance, changing V_0 to $V_0 + \sigma^2$. The history term ζ in our context (see just below Eq. 5 in [FSV21]) is $\zeta(t) = \mathbb{E}(V_t) - V_0 = \xi_0(t) - V_0$, whose contribution is $O(t^H)$ under the assumption $\xi_0(t) - V_0 = O(t^H)$. The full two-term expansion in Theorem 2.1 of [FSV21] assumes $H < 2-Y$, but the following leading-jump expansion holds on the larger range $1-Y/2 < H < 1/2$, as shown below:

Proposition 2.3 *Let $\hat{\sigma}(K,T)$ denote the implied volatility of a European call option with strike K and maturity T at time zero. For the model in (2), if $1 - \frac{1}{2}Y < H < \frac{1}{2}$, $G > 1$, $M > 1$, $|\rho| < 1$, $\xi_0(t) - V_0 = O(t^H)$ as $t \to 0$, and S is a true martingale, we have the following asymptotic behaviour for implied volatility at time zero in the Edgeworth regime:*

$$\hat{\sigma}(e^{z\sqrt{T}}, T) = \sqrt{V_0 + \sigma^2} + \frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+A_+(z) + C_-A_-(z))T^{1-\frac{1}{2}Y} + o(T^{1-\frac{1}{2}Y}) \tag{7}$$

for $z \in \mathbb{R}$ as $T \to 0$ and

$$A_\pm(z) = \frac{1}{\pi}\int_0^\infty \mathrm{Re}[e^{-iuz}e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)(\pm iu)^{Y-2}]du$$

where ϕ is the standard Normal density. We retain $|\rho| < 1$ as in the model setup of [FSV21]; the correlation endpoints are not considered here.

Proof. Put $a = 1 - Y/2$. The localisation in Appendix C allows us to replace the square-root coefficient by a bounded Lipschitz function, with an exponentially small error in the normalized put price. Let $\hat{V}$ and $\hat{X}$ denote the corresponding localized processes in that appendix, and let $X_t^{(0)} = \sqrt{V_0}(\rho W_t + \bar{\rho}B_t)$ as in Appendix B. By Itô isometry, the assumption $\xi_0(t) - V_0 = O(t^H)$ and the boundedness and Lipschitz continuity of the localized coefficient,

$$\begin{aligned}\mathbb{E}((\hat{V}_t - V_0)^2) &= O(t^{2H}), \\ \mathbb{E}(|\hat{X}_t - (X_t^{(0)} - \frac{1}{2}V_0t - L_t)|) &= O(t^{H+\frac{1}{2}} + t).\end{aligned}$$

For fixed z, the function $x \mapsto (e^{z\sqrt{t}} - e^x)^+$ is $e^{z\sqrt{t}}$-Lipschitz. After dividing by $\sqrt{t}$, the put price for the localized model therefore differs from that for the constant-volatility Gaussian-plus-Lévy model by $O(t^H)$, and the localisation

error is smaller than any power of t. The independent Brownian component of L simply changes the Gaussian variance to $V_0 + \sigma^2$.

The pure-jump calculation underlying Proposition 2.4 and Appendix C of [FSV21], which itself does not depend on H, gives the leading jump correction of order t^a with remainder $O(t^{\min(2-Y,\frac{1}{2})})$. Since $a < H < \frac{1}{2}$ and $\min(2 - Y, \frac{1}{2}) > a$, all these errors are $o(t^a)$. The leading-jump implied-volatility expansion (7) then follows by the Black–Scholes inversion argument of Lemma 2.7 in [FSV21].

The stronger condition $H < 2 - Y$ in Theorem 2.1 of [FSV21] is needed there to make the second-order jump remainder $O(t^{2-Y})$ smaller than the rough-volatility correction of order t^H. We do not retain that latter correction here, so this additional condition is unnecessary. ■

Remark 2.3 $\frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+A_+(z) + C_-A_-(z))$ is non-linear in z and does not vanish at $z = 0$ in general (see e.g. Figure 6 in [FSV21] for a plot), and C_+ and M relate to negative jumps for X and C_- and G relate to positive jumps, since we are subtracting L in (2). In particular, the rough-Heston small-time LDP for $X_t/t^{\frac{1}{2}-H}$ at speed t^{-2H} does not extend to the jump model: jumps of size $t^{\frac{1}{2}-H}$ have polynomial, rather than exponentially small, probabilities.

Remark 2.4 Note that a_1 depends on M and G as well, which do not show up in the leading order Edgeworth asymptotics in (7).

Hence the addition of the CGMY-type jump process L allows the asymptotic SPX and VIX short time smiles to have power-law skews with different H-values since the effective H as $T \to 0$ for the former is now $1 - Y/2$ (due to jumps alone, see (7)) and the latter is $H = \alpha - \frac{1}{2}$ (due to V alone), so (aside from (7) depending on V_0), we have effectively de-coupled the short-time behaviour of the SPX and VIX skews.

2.3 Exact calibration to the at-the-money skew for SPX and VIX as $T \to 0$

We let $\hat{\sigma}^{\text{mkt}}_{\text{VIX}}(K,T)$ denote the *market* implied volatility of a VIX call option with strike K and maturity T. If there exists an $\alpha = H + \frac{1}{2}$ such that $\lim_{T\to 0} \hat{\sigma}^{\text{mkt}}_{\text{VIX}}(\text{VIX}_0 e^{xT^{\frac{1}{2}-H}}, T)$ has a non-trivial limit which is differentiable in x, then we can choose V_0 and ν so that the market observed behaviour of VIX options as $T \to 0$ is consistent with (5). We can then choose parameters so that (7) is consistent with the observed behaviour of SPX options in the market, provided the parameters obtained below are admissible. Define $\hat{\sigma}^{\text{mkt}}(1,0) := \lim_{T\to 0} \hat{\sigma}^{\text{mkt}}(1,T)$. Specifically choose σ to fit this limiting at-the-money implied vol, which requires $\sigma^2 = (\hat{\sigma}^{\text{mkt}}(1,0))^2 - V_0 \geq 0$, and then solve for C_+, C_- and Y to match the observed behaviour of the at-the-money correction for European options: $\lim_{T\to 0} \frac{\hat{\sigma}^{\text{mkt}}(1,T) - \hat{\sigma}^{\text{mkt}}(1,0)}{T^{1-Y/2}}$ and the skew term: $\lim_{z\to 0} \lim_{T\to 0} \frac{\hat{\sigma}^{\text{mkt}}(e^{z\sqrt{T}},T) - \hat{\sigma}^{\text{mkt}}(1,T)}{zT^{1-Y/2}}$, assuming we can find a $Y \in (1,2)$ such that both these limits are finite (and at least one is non-zero). This leads to the following two equations for $C_\pm$:

$$\sqrt{2\pi}(C_+A_+(0) + C_-A_-(0)) = \lim_{T\to 0} \frac{\hat{\sigma}^{\text{mkt}}(1,T) - \hat{\sigma}^{\text{mkt}}(1,0)}{T^{1-Y/2}} \tag{8}$$

$$\sqrt{2\pi}(C_+A'_+(0) + C_-A'_-(0)) = \lim_{z\to 0} \lim_{T\to 0} \frac{\hat{\sigma}^{\text{mkt}}(e^{z\sqrt{T}},T) - \hat{\sigma}^{\text{mkt}}(1,T)}{zT^{1-Y/2}} \tag{9}$$

obtained by computing $\frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+A_+(z) + C_-A_-(z))$ and its derivative at $z = 0$. Any solution must also satisfy $C_+ \geq 0$ and $C_- \geq 0$; invertibility of the linear system by itself does not guarantee an admissible calibration. The last two equations are linear in C_+ and C_- and hence can be solved explicitly if the Wronskian of $A_+(z)$ and $A_-(z)$ is non-zero at $z = 0$, and we can verify that this is indeed the case since

$$\begin{aligned}
A_\pm(z) &= \frac{1}{\pi}\int_0^\infty \text{Re}[e^{-iuz}e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)(\pm iu)^{Y-2}]du \\
&= \frac{1}{\pi}\int_0^\infty \text{Re}[e^{-iuz}e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)e^{\pm\frac{1}{2}i\pi(Y-2)}u^{Y-2}]du \\
&= \frac{1}{\pi}\int_0^\infty \cos(-uz \pm \frac{1}{2}\pi(Y-2))e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)u^{Y-2}du
\end{aligned}$$

which implies that

$$\begin{aligned}
A'_\pm(z) &= \frac{1}{\pi}\int_0^\infty \text{Re}[u\sin(-uz \pm \frac{1}{2}\pi(Y-2))e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)u^{Y-2}]du \\
A_\pm(0) &= \frac{1}{\pi}\int_0^\infty \text{Re}[\cos(\frac{1}{2}\pi(Y-2))e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)u^{Y-2}]du \\
A'_\pm(0) &= \pm\frac{1}{\pi}\int_0^\infty \text{Re}[u\sin(\frac{1}{2}\pi(Y-2))e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)u^{Y-2}]du
\end{aligned}$$

i.e. $A_+(0) = A_-(0)$ and $A'_+(0) = -A'_-(0)$.

We also have to make the model consistent with the observed option-implied log-contract variance term structure (rather than, in general with jumps, the expected realised quadratic variation) up to some maturity $T^* \in (0, \Delta]$, for which

$$\begin{aligned} -2\mathbb{E}(\log \frac{S_t}{S_0}) &= \int_0^t \xi_0(u)du + a_1 t \\ \Rightarrow \quad -\frac{\partial}{\partial t} 2\mathbb{E}(\log \frac{S_t}{S_0}) &= \xi_0(t) + a_1 \\ \Rightarrow \quad -\frac{\partial}{\partial t} 2\mathbb{E}(\log \frac{S_t}{S_0})_{t=0} &= \xi_0(0) + a_1 \;=\; V_0 + a_1 \,. \end{aligned} \tag{10}$$

a_1 depends on G and M, which must be chosen so the calibrated value $\xi_0(0) = V_0$ here matches our calibrated value of V_0 from above; then (10) uniquely determines $\xi_0(t)$ for $t > 0$ as well (and this includes being consistent with the market value of VIX_0 which corresponds to $t = \Delta$), which we have implicitly used above for the VIX smile calibration. This requires checking that the inferred ξ_0 remains nonnegative and gives an admissible nonnegative solution to the square-root Volterra equation.

We now discuss the range of admissible values for a_1. Imposing the martingale condition $V(1) = 0$ we find that

$$\begin{aligned} a_1 \;=\; a_1(G,M) &= \sigma^2 - 2C_+(M^Y - (1+M)^Y + M^{Y-1}Y)\Gamma(-Y) \\ &+ 2C_-((G-1)^Y - G^Y + G^{Y-1}Y)\Gamma(-Y)\,. \end{aligned}$$

Then $a_1(1,1) > 0$ and $a_1(G,M) \to \sigma^2$ as $M, G \to \infty$, and we have the following monotonicity result:

Lemma 2.4 *$a_1(G,M)$ is continuous and nonincreasing in M and G, and is strictly decreasing in M if $C_+ > 0$ and in G if $C_- > 0$.*

Proof.

$$\frac{\partial a_1}{\partial M} \;=\; -2C_+(YM^{Y-1} - Y(1+M)^{Y-1} + Y(Y-1)M^{Y-2})\Gamma(-Y) \tag{11}$$

Now consider the function $g_M(K) = (K+M)^{Y-1}$. This function is concave so

$$(K+M)^{Y-1} \;\le\; M^{Y-1} + (Y-1)M^{Y-2}K$$

Setting $K = 1$ we see that

$$(1+M)^{Y-1} \;\le\; M^{Y-1} + (Y-1)M^{Y-2}$$

and the result follows by comparing this to (11). We use a similar argument for the term involving G. ■

Since Proposition 2.3 requires that $G, M > 1$, we see that there is an interval (σ^2, a_1^*) of admissible a_1-values, where $a_1^* = a_1(1,1) = \sigma^2 + 2(C_+(2^Y - Y - 1) + C_-(Y-1))\Gamma(-Y)$ for Y in the admissible range $Y \in (1,2)$. If the calibrated V_0-value is such that $-\frac{\partial}{\partial t} 2\mathbb{E}(\log \frac{S_t}{S_0})_{t=0} - V_0 \notin (\sigma^2, a_1^*)$ (see (10)) then it means the model is mis-specified, and this ultimately puts an upper and lower bound on the admissible ratio of the first to the zeroth order term in (5).

Of course in practice we cannot perfectly extract all the limiting quantities on the right hand side of (8) and (9) from the market since only a finite number of options are traded, and there are real-life issues with noisy data, bid-offer spreads, day count/weekend conventions etc., but one can approximate the quantities on the right with finite differences to incorporate information from options with small but non-zero maturity. These results can also be used to make smart initial guesses for conventional and deep learning-based calibration schemes.

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A Asymptotics for VIX call options

From Jensen's inequality, we know that for any $q \geq 1$ we have

$$(\mathrm{VIX}_T^2)^q \;=\; (a_1 + \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u)du)^q \;=\; (\frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\xi_T(u))du)^q \;\leq\; \frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\xi_T(u))^q du$$

and hence

$$\begin{aligned}\mathbb{E}(\mathrm{VIX}_T^{2q}) \;\leq\; \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+\xi_T(u))^q)du \;&=\; \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}(\,\mathbb{E}(a_1+V_u|\mathcal{F}_T)^q)du \\ &\leq\; \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+V_u)^q)du \qquad \text{(A-1)}\end{aligned}$$

which will be needed further down.

- Lower bound. We first note that for x fixed and any $\delta \in (0,|x|)$, $e^{xT^{\frac{1}{2}-H}} \leq 1 + (x+\delta)T^{\frac{1}{2}-H}$ for T sufficiently small. Recall that $\mathrm{VIX}_0^2 = a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du$ and we set $k_{x,\delta} := \mathrm{VIX}_0(x+\delta)$. Thus for $T = T(\delta)$ sufficiently small

$$\begin{aligned}\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0 e^{xT^{\frac{1}{2}-H}})_+) \;&\geq\; \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+(x+\delta)T^{\frac{1}{2}-H}))_+) \\ &=\; T^{\frac{1}{2}-H}\mathbb{E}((\frac{\mathrm{VIX}_T-\mathrm{VIX}_0}{T^{\frac{1}{2}-H}} - k_{x,\delta})_+) \\ &\geq\; \delta T^{\frac{1}{2}-H}\mathbb{E}(1_{\frac{\mathrm{VIX}_T-\mathrm{VIX}_0}{T^{\frac{1}{2}-H}} > k_{x,\delta}+\delta}) \\ &=\; \delta T^{\frac{1}{2}-H}\mathbb{P}(\mathrm{VIX}_T > \mathrm{VIX}_0 + T^{\frac{1}{2}-H}(k_{x,\delta}+\delta)) \\ &=\; \delta T^{\frac{1}{2}-H}\,\mathbb{P}(\mathrm{VIX}_T^2 > \mathrm{VIX}_0^2 + 2\mathrm{VIX}_0\,(k_{x,\delta}+\delta)T^{\frac{1}{2}-H} + (k_{x,\delta}+\delta)^2 T^{1-2H})\,.\end{aligned}$$

  But for $T = T(\delta)$ sufficiently small, the right hand side here is greater than or equal to

$$\begin{aligned}&\delta T^{\frac{1}{2}-H}\,\mathbb{P}(\mathrm{VIX}_T^2 - \mathrm{VIX}_0^2 > 2\mathrm{VIX}_0\,(k_{x,\delta}+2\delta)T^{\frac{1}{2}-H}) \\ =\;& \delta T^{\frac{1}{2}-H}\,\mathbb{P}(\frac{1}{\Delta}\int_T^{T+\Delta}\xi_T(u)du - \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du > 2\mathrm{VIX}_0\,(k_{x,\delta}+2\delta)T^{\frac{1}{2}-H})\,.\end{aligned}$$

  Then using the LDP and the continuity of J we see that

$$\begin{aligned}\liminf_{T\to 0} T^{2H}\log\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,e^{xT^{\frac{1}{2}-H}})_+) \;&\geq\; -J(2\mathrm{VIX}_0(k_{x,\delta}+2\delta)) \\ &=\; -J(2\mathrm{VIX}_0^2 x + 2\delta\mathrm{VIX}_0^2 + 4\delta\mathrm{VIX}_0)\,.\end{aligned}$$

  We then let $\delta \to 0$ and again use the continuity of the rate function $J(x)$ to obtain the required lower bound.

- Upper bound. From Hölder's inequality, we note that for $q > 1$

$$\begin{aligned}\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\, e^{xT^{\frac{1}{2}-H}})_+) &\leq \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+)\\ &= \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+ 1_{\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H})})\\ &\leq \mathbb{E}[(\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+^q]^{\frac{1}{q}}\, \mathbb{E}(1_{\mathrm{VIX}_T \geq \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H})})^{1-\frac{1}{q}}.\end{aligned}$$

Thus

$$\begin{aligned}& T^{2H}\log\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))_+)\\ \leq\ & \frac{T^{2H}}{q}\log\mathbb{E}[(\mathrm{VIX}_T - \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))_+^q] \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))\\ \leq\ & \frac{T^{2H}}{q}\log\mathbb{E}(\mathrm{VIX}_T^q) \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))\\ \leq\ & \frac{T^{2H}}{q}\log(\mathbb{E}(\mathrm{VIX}_T^{2q})^{\frac{1}{2}}) \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))\\ \leq\ & \frac{T^{2H}}{q}\frac{1}{2}\log(\frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+V_u)^q)du) \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+xT^{\frac{1}{2}-H})^2)\\ & \text{(by (A-1))}\\ \leq\ & \frac{T^{2H}}{q}\frac{1}{2}\log(\frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\mathbb{E}(V_u^q)^{\frac{1}{q}})^q du) \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+2xT^{\frac{1}{2}-H}))\\ & \text{(using Minkowski applied to } \mathbb{E}((a_1+V_u)^q))\\ \leq\ & \frac{T^{2H}}{q}\frac{1}{2}\log(a_1+c_q^{\frac{1}{q}})^q \,+\, T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+2xT^{\frac{1}{2}-H}))\end{aligned}$$

where for each $q > 1$ we fix $T_0 > 0$ and take $c_q := \sup_{0\leq u\leq \Delta+T_0}\mathbb{E}(V_u^q) < \infty$, uniformly for $0 < T \leq T_0$, using Lemma C.1 in [FGS21b] in the final line. Thus $T^{2H}\log(a_1 + c_q^{1/q}) \to 0$ as $T \to 0$. Letting $T \to 0$ in the final line and using the LDP and the continuity of J, and then letting $q \to \infty$, we see that

$$\limsup_{T\to 0} T^{2H}\log\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))_+) \quad\leq\quad -J(2\mathrm{VIX}_0^2 x)\,.$$

B Bivariate weak convergence

Proposition B.1 *Let $\widetilde{X}_t := X_t + L_t$ denote the continuous rough-Heston log-price component and let $X_t^{(0)} := \sqrt{V_0}(\rho W_t + \bar\rho B_t)$. Then $(\widetilde{X}_t/\sqrt{t}, (V_t - \xi_0(t))/t^H)$ and $(X_t^{(0)}/\sqrt{t}, (V_t - \xi_0(t))/t^H)$ both tend weakly to the same bivariate Gaussian as $t \to 0$.*

Proof. By the Volterra equation for V and Itô isometry, using $\mathbb{E}(V_s) = \xi_0(s)$,

$$\mathbb{E}((V_t - \xi_0(t))^2) \quad=\quad \nu^2 c_\alpha^2 \int_0^t (t-s)^{2\alpha-2}\xi_0(s)ds \leq Ct^{2H}$$

for all sufficiently small t, since ξ_0 is continuous. Thus

$$\sup_{0\leq s\leq t}\mathbb{E}(|V_s - V_0|) \quad\leq\quad \sup_{0\leq s\leq t}|\xi_0(s) - V_0| + Ct^H \to 0.$$

Since $\widetilde{X}_t = -\frac{1}{2}\int_0^t V_s ds + \int_0^t \sqrt{V_s}(\rho dW_s + \bar\rho dB_s)$, the Itô isometry and $|\sqrt{v} - \sqrt{V_0}|^2 \leq |v - V_0|$ for $v \geq 0$ imply

$$\mathbb{E}((\frac{1}{\sqrt{t}}\int_0^t(\sqrt{V_s} - \sqrt{V_0})(\rho dW_s + \bar\rho dB_s))^2) \quad\leq\quad \sup_{0\leq s\leq t}\mathbb{E}(|V_s - V_0|) \to 0.$$

Also, $\mathbb{E}((t^{-1/2}\int_0^t V_s ds)^2) \leq \int_0^t \mathbb{E}(V_s^2)ds = O(t)$, since the preceding second-moment bound makes $\mathbb{E}(V_s^2)$ locally bounded. Thus $\frac{\widetilde{X}_t - X_t^{(0)}}{\sqrt{t}} \to 0$ in L^2. Similarly, by Brownian scaling, define $\widetilde{W}_r^\varepsilon := \varepsilon^{-\frac{1}{2}}W_{\varepsilon r}$, $\widetilde{B}_r^\varepsilon := \varepsilon^{-\frac{1}{2}}B_{\varepsilon r}$. Then

$$\begin{aligned}\frac{X_{\varepsilon t}^{(0)}}{\sqrt{\varepsilon}} &= \sqrt{V_0}(\rho\widetilde{W}_t^\varepsilon + \bar\rho\widetilde{B}_t^\varepsilon),\\ \frac{V_{\varepsilon t} - \xi_0(\varepsilon t)}{\varepsilon^H} &= \nu c_\alpha\int_0^t (t-r)^{\alpha-1}\sqrt{V_{\varepsilon r}}d\widetilde{W}_r^\varepsilon.\end{aligned}$$

Since $\sup_{0\le r\le t}\mathbb{E}(|V_{\varepsilon r}-V_0|)\to 0$ and $(t-r)^{\alpha-1}$ is square-integrable on this interval, Itô isometry gives convergence of the stochastic integral in L^2. We therefore obtain

$$(\frac{X^{(0)}_{\varepsilon t}}{\sqrt{\varepsilon}},\frac{V_{\varepsilon t}-\xi_0(\varepsilon t)}{\varepsilon^H}) \quad \stackrel{w}{\to} \quad (\sqrt{V_0}(\rho W_t+\bar\rho B_t),\,\nu\sqrt{V_0}c_\alpha\int_0^t(t-r)^{\alpha-1}dW_r).$$

The limiting vector is centred Gaussian with moment generating function $e^{V_0(\frac{1}{2}p^2t+p\rho\nu u\frac{t^\alpha}{\Gamma(\alpha+1)}+\frac{1}{2}\nu^2u^2\frac{t^{2\alpha-1}}{\Gamma(\alpha)^2(2\alpha-1)})}$. By Slutsky's theorem and the preceding L^2 estimate, the same weak limit holds with $X^{(0)}_t$ replaced by $\widetilde{X}_t$. Note that the additional independent Brownian component of L contributes σ^2 to the variance of the weak limit for the full log price $X_t/\sqrt{t}$, so the corresponding assertion for X_t itself would be false when $\sigma>0$. ■

C Extending to unbounded volatility

This appendix is a minor variation of Section 2.1 in [FSV21] adapted for our purposes here. We first let $\underline{V}_t := \min_{0\le s\le t}V_s$ and $\bar V_t := \max_{0\le s\le t}V_s$, and we consider the stochastic Volterra system

$$\hat V_t \;=\; \xi_0(t)+c_\alpha\nu\int_0^t(t-s)^{H-\frac{1}{2}}\sigma(\hat V_s)dW_s$$

where $\sigma\in C^2_b$ as in [FSV21] and bounded away from zero with $\sigma(y)=\sqrt{y}$ for $y\in[a,b]$ for some $a\in(0,V_0)$ and $b>V_0$, so in particular σ is bounded and Lipschitz. Then $V_t=\hat V_t$ for $t\in[0,\tau_a\wedge\tau_b)$ where $\tau_a=\inf\{t:V_t=a\}$ (and similarly for τ_b) for any $a\in(0,V_0)$ since we have pathwise uniqueness for $\hat V$ (see section 4.2.3 in [JP20] and discussion in section 2 in [FGS21]) and from Proposition 4.3 in [JP20] (or Theorem 3.11 in [Zha08]), we know that $\hat V_{t(.)}$ satisfies the large deviation principle on $C_{V_0}[0,1]$ as $t\to0$ with speed $1/t^{2H}$ and rate function

$$I(\phi) \;=\; \frac{1}{2}\int_0^1(\frac{D^\alpha(\phi(.)-\phi(0))(t)}{\nu\,\sigma(\phi(t))})^2dt$$

if $\phi\in I_{V_0}^{H+\frac{1}{2}}(L^1)$, and $I(\phi)=\infty$ otherwise. Now let $A:=\{\underline{V}_t>a\}\cap\{\bar V_t<b\}$ and set $\hat S_t=e^{\hat X_t}$, where

$$d\hat X_t \;=\; -\frac{1}{2}\sigma(\hat V_t)^2dt+\sigma(\hat V_t)(\rho dW_t+\bar\rho dB_t)-dL_t$$

and $\hat X_0=0$ i.e. the same stock price process (defined on the same probability space) but with V_t replaced with $\hat V_t$. Then for all $\varepsilon>0$ we have

$$\begin{aligned}\mathbb{E}((e^{z\sqrt{t}}-S_t)^+) \;&=\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+1_A)+\mathbb{E}((e^{z\sqrt{t}}-S_t)^+1_{A^c})\\ &\le\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)+e^{z\sqrt{t}}\mathbb{P}(A^c)\\ &\le\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)+e^{z\sqrt{t}}e^{-\frac{1}{t^{2H}}(-\varepsilon+J(a,b))}\end{aligned}$$

for t sufficiently small, where $J(a,b)=\inf_{\phi\in C_{V_0}([0,1]):\underline{\phi}(1)\le a\text{ or }\bar\phi(1)\ge b}I(\phi)$, and we have used the upper bound implied by the aforementioned LDP in the final line. For this to be useful, we need to check that $J(a,b)>0$. To this end, let $\mathcal{I}:H^1_0\to C_{V_0}([0,1])$ be the Itô map which takes h to the solution to $\phi(t)=V_0+\int_0^t c_\alpha\nu(t-s)^{H-\frac{1}{2}}\dot h_s\sigma(\phi(s))ds$ (recall this solution is unique since σ is bounded and Lipschitz). Then

$$\begin{aligned}|\mathcal{I}(h)(t)-V_0| \;=\; |\phi(t)-V_0| \;=\; |\int_0^t c_\alpha\nu(t-s)^{H-\frac{1}{2}}\sigma(\phi(s))\dot h_sds| \;&\le\; \|\sigma\|_\infty\int_0^t c_\alpha\nu(t-s)^{H-\frac{1}{2}}|\dot h_s|ds\\ &\le\; const.\times\frac{1}{2}(\int_0^t\dot h_s^2ds)^{\frac{1}{2}}\\ &\le\; const.\times I(\phi)^{\frac{1}{2}}\end{aligned}$$

where we have used Cauchy-Schwarz in the penultimate inequality. If $J(a,b)=0$ then for all $\varepsilon>0$ there exists a $\phi_\varepsilon\in C_{V_0}([0,1])$ with $\underline{\phi}_\varepsilon(1)\le a$ or $\bar\phi_\varepsilon(1)\ge b$ and $I(\phi_\varepsilon)\le\varepsilon$, which (from the bound immediately above) implies that the sup norm of $\phi_\varepsilon-V_0$ is less than $(b-V_0)\wedge(V_0-a)$ for ε sufficiently small, which is a contradiction. Thus $J(a,b)>0$.

Similarly

$$\begin{aligned}\mathbb{E}((e^{z\sqrt{t}}-S_t)^+) \;\ge\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+1_A) \;&=\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)-\mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+1_{A^c})\\ &\ge\; \mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)-e^{z\sqrt{t}}e^{-\frac{1}{t^{2H}}(J(a,b)-\varepsilon)}.\end{aligned}$$

Thus

$$\begin{aligned}\frac{1}{\sqrt{t}}\mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)-\frac{e^{z\sqrt{t}}}{\sqrt{t}}e^{-\frac{1}{t^{2H}}(J(a,b)-\varepsilon)} \;&\le\; \frac{1}{\sqrt{t}}\mathbb{E}((e^{z\sqrt{t}}-S_t)^+)\\ &\le\; \frac{1}{\sqrt{t}}\mathbb{E}((e^{z\sqrt{t}}-\hat S_t)^+)+\frac{e^{z\sqrt{t}}}{\sqrt{t}}e^{-\frac{1}{t^{2H}}(J(a,b)-\varepsilon)}.\end{aligned}$$

For the localized model $\hat{S}$, the quantitative comparison with the Gaussian-plus-Lévy model in the proof of Proposition 2.3 gives the leading jump expansion with an $o(t^{1-Y/2})$ remainder. The bounds above show that the difference between its normalized put price and that of S is exponentially small in t^{-2H}, so this error is also $o(t^{1-Y/2})$. Consequently the same leading-jump expansion holds for S.